

Total No. of Questions : 8]

SEAT No. :

PD4902

[Total No. of Pages : 2

[6404]-460

B.E. (Computer Engineering)

HONORS IN DATA SCIENCE

**Artificial Intelligence for Big Data Analytics
(2019 Pattern) (Semester - VIII) (410503)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is artificial neural networks and state its applications? [6]
b) Explain how to avoid overfitting in Neural Networks. [6]
c) What is a recurrent neural network and how it differs from feed forward neural networks? [8]

OR

- Q2)** a) Explain features of back propagation. [6]
b) What is gradient descent and explain its types? [6]
c) What is perceptron and explain Components of a Perceptron. [8]

- Q3)** a) What are the main components of Hadoop? [4]
b) Explain types of nodes in Hadoop. [6]
c) What are the features of HDFS explain in detail? [8]

OR

- Q4)** a) What are the challenges of Map Reduce. [4]
b) What is of Map Reduce? What are the steps in Map Reduce? [6]
c) What is Hadoop Streaming and explain how it works. [8]

P.T.O.

- Q5)** a) What are the key components of data warehouse? [6]
b) What is data ingestion and explain its types. [6]
c) Write a short note on: Data mining. [4]

OR

- Q6)** a) Explain Scalable Machine Learning on Big Data using Spark. [6]
b) Explain the benefits of data ingestion in detail. [6]
c) What is Hive and why do we need it? [4]

- Q7)** a) What are the applications of NLP? [6]
b) What is pre-processing in computer vision? [6]
c) Explain feature extraction in NLP. [4]

OR

- Q8)** a) Explain in details Text pre-processing in NLP. [6]
b) How to Apply NLP Techniques? [6]
c) What are the applications of computer vision? [4]

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